

DATA STRUCTURE

LAB PROGRAMS

9. Write a C program to search a number using Binary Search method #include <stdio.h>

```
int main() {
    int arr[100], n, key;
    int low, high, mid, found = 0, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter sorted array elements:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while(low <= high) {
        mid = (low + high) / 2;

        if(arr[mid] == key) {
```

```
        found = 1;
        printf("Element found at position %d", mid + 1);
        break;
    }
    else if(arr[mid] < key)
        low = mid + 1;
    else
        high = mid - 1;
}

if(found == 0)
    printf("Element not found");

return 0;
}
```

```
Enter number of elements: 2
Enter sorted array elements:
3 4
Enter element to search: 4
Element found at position 2
-----
Process exited after 6.355 seconds with return value 0
Press any key to continue . . .
```